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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.3

$$\int \frac{4x^2-6x+1}{2x^3-x^2}dx = \int \frac{4x^2-6x+1}{x^2(2x-1)}dx$$
$$\frac{4x^2-6x+1}{x^2(2x-1)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{2x-1}$$
$$A = \left.\frac{4x^2-6x+1}{2x-1}\right|_{x=0} = -1$$
$$B = \frac{4x^2-6x+1}{x^2(2x-1)}+\frac{1}{x^2} = \frac{4x^2-6x+1+(2x-1)}{x^2(2x-1)} = \frac{4x^2-4x}{x^2(2x-1)} = \frac{4x-4}{x(2x-1)}$$
$$\Rightarrow B = \left.\frac{4x-4}{2x-1}\right|_{x=0} = 4$$
$$C = \left.\frac{4x^2-6x+1}{x^2}\right|_{x=\frac{1}{2}} = -4$$
$$\Rightarrow -\int \frac{dx}{x^2} + 4\int \frac{dx}{x} - 4\int \frac{dx}{2x-1}$$
$$= \frac{1}{x} + 4\ln|x| - 2\ln|2x-1| + c$$

References